The Measurement Revolution



Executive Summary

The Measurement Revolution is a direct result of another revolution – the PC revolution – that has affected every aspect of our lives. Technology being developed by the multibillion dollar computing industry is flooding every other industry and application. The Measurement Revolution is the story of how the PC has remodeled measurement and automation. It started with the general-purpose interface bus (GPIB), a device with which electronic instruments could be connected to and controlled from computers. GPIB was the first marriage of the computer and measurement devices. Once computers were connected to instruments, engineers and scientists saw the opportunity to exploit the computer in their measurement systems. Software, such as LabVIEW™, made connecting instruments to a computer as easy as drawing a flow chart – a familiar task for engineers. Then, with the introduction of data acquisition (DAQ), measurement devices that plugged directly in the computer, engineers and scientists began replacing older, vendor-defined instruments with lower cost, modular measurement components and software. And now, with so many generic measurement components, such as data acquisition boards, image acquisition boards, and computer-based instruments, any measurement system with powerful software can take advantage of the computer. The result is the digital measurement system.

By building digital measurement systems, which take advantage of the latest computer technologies, the developer reaps the benefits of greater system performance and lower total system cost. As the PC becomes faster and faster, a measurement system that fully integrates the PC continues to see increased performance. Furthermore, the cost of upgraded performance is minimal. Digital measurement systems typically cost one half to one tenth the price of a stand-alone system and offer greater performance.

The key to a digital measurement system is highly integrated, componentized software and hardware. The measurement hardware is typically just a digitizer that connects to the computer through a high-speed interface bus. As long as the digitizer can measure certain frequencies within a specified accuracy, any measurement is possible. The software is the differentiating component of the digital measurement system. By using highly productive development tools, developers can easily connect to measurement components for high-speed, high-accuracy measurements. They also can quickly add highly advanced analysis routines, publish data via the Internet, and communicate with corporate databases. With modular, hardware-independent code, developers are not tied to today's technology. If a faster computer or better digitizer comes along, the developer of a digital measurement system can take advantage of minimal investments in hardware and no changes to software code.

Developers can take the digital measurement system to a new level with truly hardware-independent software. Then, a developer can take the system anywhere. A digital measurement system could be running on an embedded computer, on a plug-in board in real-time, or embedded in a distributed module. Anything is possible with a platform as flexible and open as a digital measurement system.

Measurement Revolution

The computer, along with the technological innovations that have driven the computer industry, has literally revolutionized markets and business processes by which companies conduct their business. And while measurement technology has improved in the same time period, the computer itself has impacted how we measure today more than any other technology. Even the core technology advances that have radically improved sensors, digitizers, and signal processing, can be traced to related advances in computer technology.

The past 20 years, the computer has been the core of the Measurement Revolution. The Measurement Revolution has transformed test and measurement and industrial automation from loosely coupled, and often incompatible, stand-alone instruments to tightly integrated, high-performance measurement and automation systems. The Measurement Revolution is actually long overdue. Considering the processing power of today's computers (and their incredibly low cost), why haven't the test and measurement or industrial automation industries produced the same large scale improvements in performance and reductions in price? These industries are just now beginning to see the large-scale benefits of computer-based measurement and automation.

Performance is by far the most obvious benefit of more effective computer use. The computer itself is more than 1,000 times faster than it was 10 years ago. Measurement systems, however, that do not fully integrate the computer have not improved at the same pace over the past decade. The reason is simple – stand-alone instruments cannot take full advantage of the processing power, nor the enterprise-wide connectivity of today's personal computer.

Cost is the other compelling side effect of the Measurement Revolution. Typically, a computer-based measurement and automation system will cost one half to one tenth that of its stand-alone equivalent. And, not only are the computers themselves extremely low in cost, the cost to upgrade to a higher performance model in the future is either negligible or zero. Developers gain the same cost savings with today's networking capabilities.

The Measurement Revolution clearly has redefined how measurements are made and has made it possible to build high-performance, tightly integrated systems at a lower cost. These systems, now possible due to the Measurement Revolution, are called "digital measurement systems."

The Evolution of Virtual Instrumentation

The digital measurement system is not a brand new concept – it is the next evolution of *virtual instrumentation.* More than 10 years ago, the LabVIEW community pioneered the use of virtual instrumentation. At that time, the term suggested that software – not hardware – should define the operation of an instrument or other measurement device. With virtual instrumentation, anyone could easily connect disparate instruments together across many different communication interfaces and program them all in the same manner. The development productivity gains were enormous. Virtual instrumentation essentially has made it possible for anyone, regardless of their computer expertise to develop sophisticated instrumentation systems. Now, the community that established virtual instrumentation is being joined by the large majority of engineers and scientists who demand the higher system throughput, better connectivity, and lower cost achieved with the digital measurement system.

The digital measurement system now encompasses more than just integrating GPIB instruments and plug-in data acquisition (DAQ) boards. Today, measurements can be made through distributed networks, high-speed computer buses, and directly in the sensors themselves. The introduction of so many new measurement components and the evolution of software development tools that can control and communicate with these components make all this possible. A typical measurement system today consists of electronic measurement devices, such as oscilloscopes, digital multimeters, analyzers, DAQ devices, image acquisition and machine vision devices, motion control, and the software to integrate all of these devices. Developers can connect today's systems to networks, publish data via Web browsers, and communicate via modems and RF links.

Key Elements of the Digital Measurement System

Computer

The computer is the core of the digital measurement system. By taking advantage of every element of the computer, the digital measurement system delivers more than any traditional measurement system ever could.

The processor brings tremendous computing power to the system. Computations needed in a measurement system, such as Fast Fourier Transform (FFTs) and other signal processing routines, can be extremely complex, but today's processors deliver answers quicker than we ever thought possible. And, they continue to get better (see Table 1).

Table 1. Execution Time for 1024-Point FFT Using LabVIEW

Microprocessor	Execution Time
500 MHz Pentium III	0.49 ms
400 MHz G3	0.52 ms
450 MHz Pentium III	0.54 ms
350 MHz G3	0.58 ms
300 MHz Pentium II	0.83 ms

Computer memory further enhances the capabilities of the digital measurement system. The combination of powerful processors and the ever increasing amount of RAM present in today's computers means that developers can now achieve many tasks once performed with a simple software routine with complicated hardware configurations. Furthermore, the size of hard drives and the speed at which data can be streamed from measurement devices to the computer continues to improve. Computers, which have hard disks up to 10 GB, rival the storage power of any data logger.

The computer in the digital measurement system can take on many forms (see Figure 2). In fact, many measurement platforms are themselves computers. For example in the 1980s, the VXI specification, which is based on the VMEbus, became the first measurement platform based solely on computing technology. Since then, further advances in the industry have made it possible to create the next generation measurement platform, – PXI™ a smaller, lower cost, higher performance platform based on CompactPCI bus. No matter which platform is chosen as the core of the digital measurement system, a standard desktop

PC, a laptop PC, or a powerful PXI controller, the benefits are the same – extraordinary amounts of computing power at virtually no cost.

High-Speed I/O Interfaces

For a digital measurement system to be effective, it must quickly transfer data from the measurement component to the computer, where most of the analysis and measurement calculations are completed. To do this, most devices in a digital measurement system are *memory-mapped* devices.

Message-Based Devices

Many measurement systems today connect instruments to computers using *message-based* architecture. GPIB, or IEEE 488, is a form of message-based I/O. In this string-based model, the instrument and the computer pass commands and data between them. Because the measurement device, or instrument, must embed some intelligence to parse and interpret the incoming messages, the transfer of data typically takes several steps, many of which could be avoided if the computer had direct access to the internal instrument registers. Although high-speed or HS488™ increases the data throughput of the IEEE 488 bus, GPIB-controlled instruments use an I/O approach that gives the measurement system little control.

Memory-Mapped Devices

Devices that measure data directly are called memory-mapped devices. By using memory-mapped devices, measurements are collected into the system directly through a high-speed bus, such as PCI, IEEE 1394, USB, or VXI. The advantage of a memory-mapped device is that the instrument or measurement component needs little intelligence. Thus, the very efficient PC can stream data directly into the computer at extremely high rates.

Table 2. Throughput Comparison among Platforms

GPIB	USB	ISA	1394	PCI	CompactPCI/PXI
1 Mbytes/s (3-wire) 8 Mbytes/s (HS488)	1.2 Mbytes/s	1-2 Mbytes/s	43 Mbytes/s >100 Mbytes/s (being defined)	132 Mbytes/s	132 Mbytes/s

Examples of memory-mapped devices are DAQ boards, computer-based instruments, image acquisition boards, and more. However, not all memory-mapped devices take advantage of the high-speed interface bus on which they were built. The driver software and the hardware on the board must tightly integrate the measurement capability with the I/O bus if the measurement device is to thoroughly reap the benefits of a high-speed interface bus, such as PCI.

In the case of PCI, with bandwidth up to 132 Mbytes/s, the memory-mapped devices in the system must integrate tightly with the PCIbus to take full advantage of the bandwidth it offers. PCI and PXI/CompactPCI measurement devices, such as National Instruments DAQ boards and computer-based instruments, have bus mastering and scatter-gather DMA

technology built into the system so that measurements are not lost in the high-speed transfer from the device to the PC. Furthermore, this technology is seamless because it is built into the software and hardware in the system.

Network Connectivity

The Internet and the network technology on which it is built are ushering in a new age of data sharing. Publishing information in real time from the measurement system to anywhere else in the world was once nearly impossible. Now, it happens in almost every system. However, developing measurement applications that take advantage of network technology is not always easy (Figure 1).

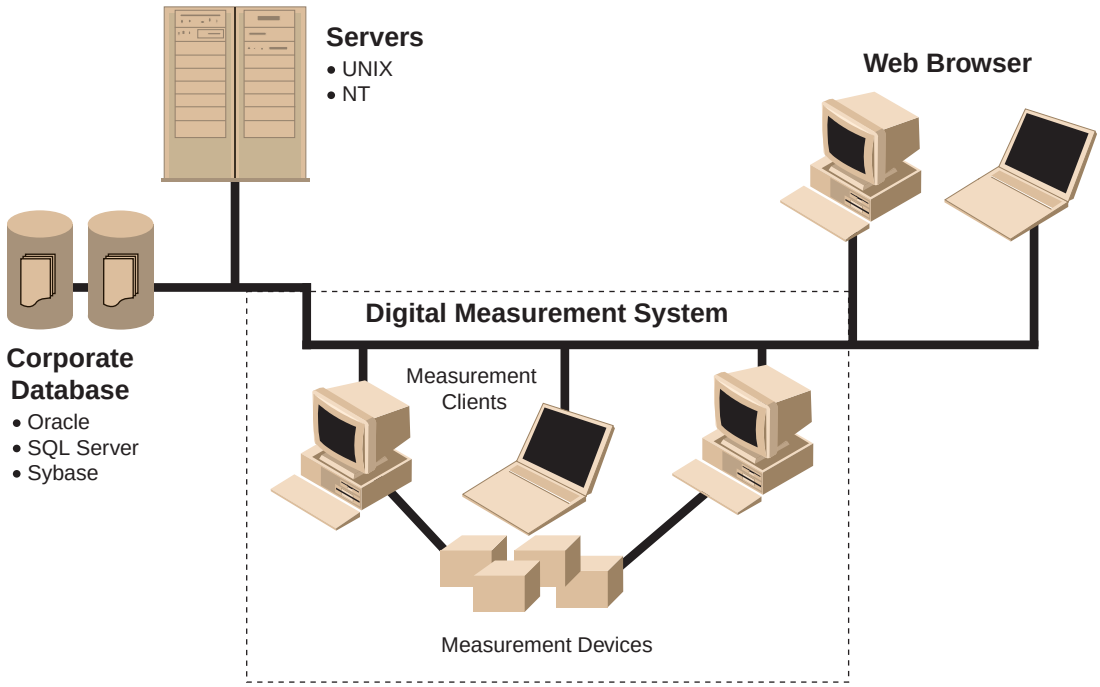


Figure 1. The Networking Aspect of a Digital Measurement System

In a digital measurement system, developers can easily plug networking components into their development environments. With modular software, developers can separate the data sharing aspect of the system from the measurement portion. Whether the system requires collecting data through a network, publishing data via the Internet, or distributing the system among several machines, software technology makes developing networked applications elementary.

Networked Data Collection

In the case of collecting data through an Ethernet connection via an Ethernet-based measurement component, the digital measurement system treats data collection exactly as it would if the measurement component were connected to the PC via any other bus. Essentially, the development software is identical, and the communication protocol is handled in driver software that is transparent to the developer.

For example, an application which collects data from a PCI-based DAQ board and writes to a file uses the same code in LabVIEW as collecting data from an Ethernet-based DAQ box (Figure 2).

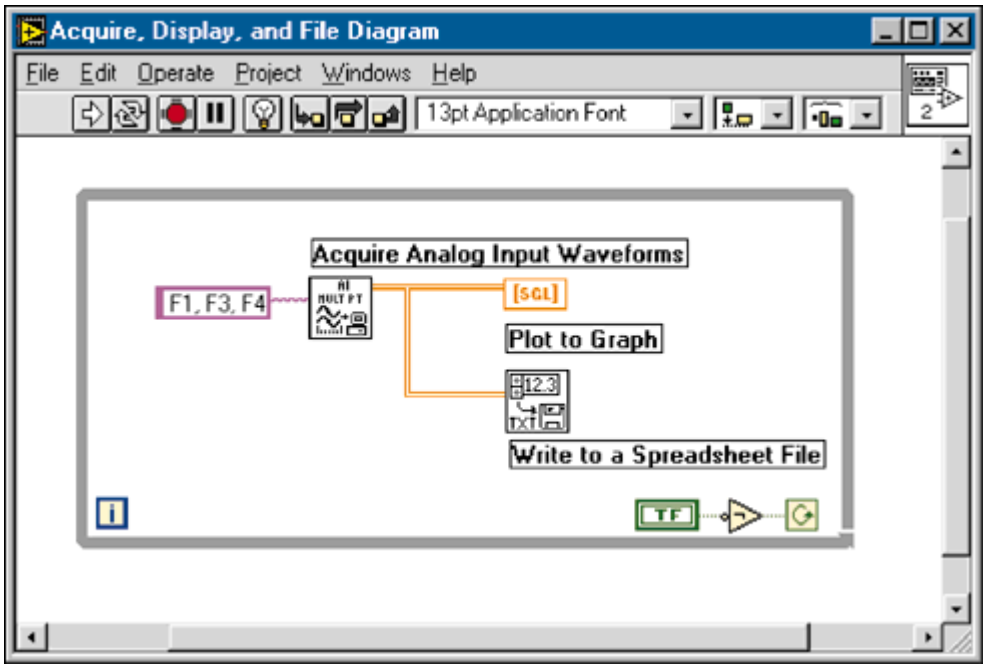


Figure 2. LabVIEW Code for Acquiring Data from a DAQ Board Connected via Any Bus

Data Publishing via the Internet

In terms of a digital measurement system, developers need to publish data via the Internet when they need to broadcast acquired and analyzed information to others for interpretation. For example, many manufacturing facilities publish data to corporate databases so that they can evaluate information coming from all aspects of the plant. Again, it is the software that makes this possible. Software standards, such as OLE for process control (OPC), define a protocol for communicating data via networks inside large facilities. However, if the digital measurement system only needs to publish data to specific applications or users, other software standards, such as DataSocket™, make Internet communication as easy as file I/O.

Distributed Systems

Distributing the digital measurement system across many devices is another feature that the Internet provides. Perhaps a particular task is time-critical and needs a dedicated processor. With intuitive software and the power of the Internet, the digital measurement system can distribute the time critical task to one device and house the rest of the application in another computer.

Measurement Components

Another vital element of the digital measurement system is the measurement component. In the digital measurement system, all of the intelligence and computing power of the system shifts from the instrument to the computer. Consequently, the only function of measurement components is to digitize the measurement and transfer the data to the PC.

Measurement components are essentially digitizers. Any measurement in a system can be characterized by knowing the speed and resolution at which the measurement will be made. Thus, sample rate and resolution are the basis for deciding which type of digitizer is needed for the digital measurement system. Figure 3 shows the digitizing range of PC-based measurement components alongside that of traditional instruments. As you can see, measurement components now solve most measurement needs – typically at a fraction of the cost.

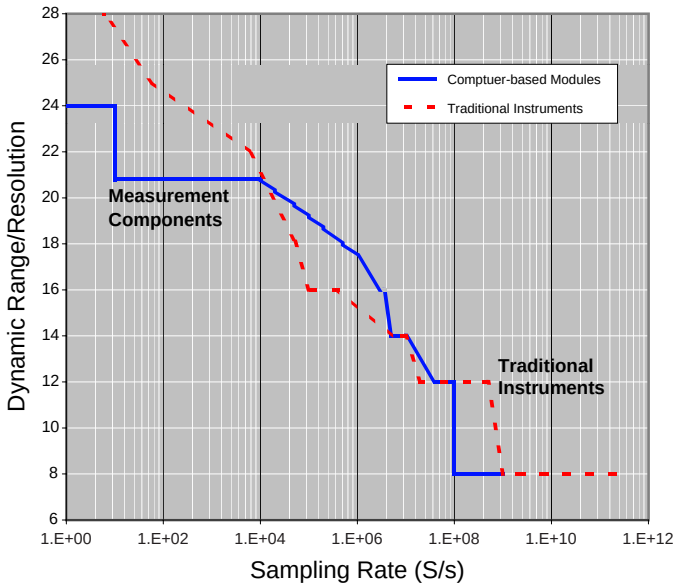


Figure 3. Sample Rate and Resolution of Measurement Components vs. Traditional Stand-Alone Instruments

Because of their elemental nature, measurement components bring an incredible level of flexibility to the system. In fact, new technologies in digitizers now offer even more flexibility. As a result of patented digital signal processing techniques, new flexible resolution technology creates digitizers that can vary the sample rate and resolution at which measurements are made. The flexible resolution technology consists of a specialized filter, a flash analog-to-digital converter, a digital-to-analog converter, and a DSP for decimation and linearization. The flexible resolution technology, which is the essence of the NI 5911 digitizer, has different modes. In oscilloscope mode, you can digitize signals with bandwidths between 4 and 100 MHz. When digitizing signals with bandwidths less than 4 MHz, you can use the flexible resolution mode to extend the effective resolution up to 20 bits.

For several years, developers in the PC-based DAQ community have used the concept of measurement components – evaluated based on sampling rate and resolution – to simplify the selection of DAQ boards. Measurement components, such as DAQ boards and computer-based instruments, simplify the design of a measurement system. With tightly integrated software, a developer can transform a measurement component into a solution for virtually any application.

Software

The software ties the entire digital measurement system together (see Figure 4). Notice that the software architecture is organized into modular components. Other elements that characterize the software in a digital measurement system are tight integration and high development productivity.

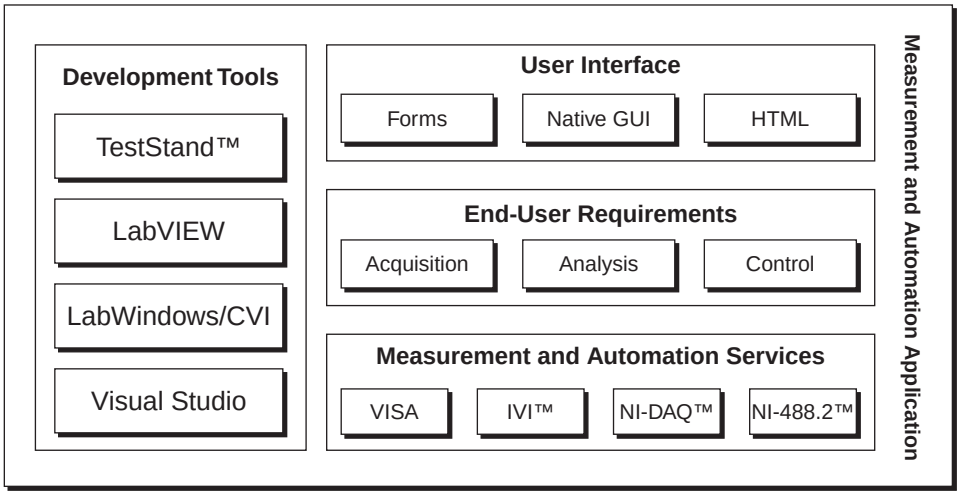


Figure 4. The Measurement and Automation Software Architecture

Modular Software Components

The software in a digital measurement system must be modular. Software development tools such as LabVIEW, give the developer the flexibility to create different software for different tasks. Then, a developer can assemble those software components to form a complicated measurement system. There are several benefits of developing with software components versus creating one large system application:

- Hardware independence – The measurement and automation services layer is a set of drivers whose underlying software architecture is very similar. Thus, developers can create hardware with related characteristics using the same high-level commands. Because the hardware variations are taken care of through the measurement and automation services layer, developers can mix and match in the system without changing software.

- Code reuse – Suppose a new application requires many of the same tasks as an existing system. If the first system was developed with componentized software modules, the developer can easily reuse its software in the new application.
- Application extension – To protect the investment in software, the measurement system must change and grow with technology. By using modular software components, developers can quickly and easily add new technology to an application. For example, as exciting new data sharing techniques emerge in the future, the data sharing portion of the measurement application can be exchanged for a new, better performing technology.

Tight Integration

To take advantage of the continued advancements in computer technology, the software and hardware in a digital measurement system must be fully integrated. Consider the components of a digital measurement system, including the software components, as pieces of a puzzle; and the developer can put the pieces together to accomplish any task. Each component must be independent, as demonstrated above, but each component also must be integrated to maximize efficiency.

Development Productivity

Easy-to-use application software greatly simplifies development of the digital measurement system.

- Acquisition – Built-in libraries expose the measurement component through the measurement and automation services software layer, so the developer can quickly and intuitively acquire information into the computer.
- Analysis – Powerful signal processing and control algorithms also are inherent to development software. By using the processing power in the PC, routines such as noise reduction, spectral leakage reduction, network analysis, signal detection and separation, interpolation and extrapolation, differentiation and integration, and trajectory estimation require only milliseconds.
- Presentation – Graphical user interface (GUI) controls, such as knobs, graphs, and numeric displays present information more clearly. Furthermore, libraries for file I/O and data sharing give the system the flexibility to present information in any conceivable way.

Essentially, the functionality now found in traditional, stand-alone system is easily reproduced and improved using the PC, flexible measurement components, and powerful development software.

The Scalability of the Digital Measurement System

One reason for the Measurement Revolution has been the billions of dollars invested by the computing and electronics industries to build microprocessors and computing devices that are better, faster, and cheaper. These computing advancements, coupled with innovative measurement technology, are fueling the Measurement Revolution every day. But, the revolution is moving quickly, and the test and measurement industry must keep up!

Platform Scalability

The Measurement Revolution results in a digital measurement system solution that is designed to anticipate technology transitions and scale for the future. Since the widespread popularity of GPIB in the 70s and 80s, software standards, such as NI-488™, have helped preserve system scalability. Although 25 years ago GPIB interfaces did not exist for computer buses such as PCI, PCMCIA, and PXI/CompactPCI, our customers saved hours of software development when upgrading to new buses because National Instruments developed the NI-488 software standard. Another example is operating system (OS) platform scalability. Nobody knows what new OS is on the horizon. Will Windows continue to dominate? What about Mac OS X, or maybe Linux? Developing digital measurement systems with development tools such as LabVIEW protects developers from having to rewrite valuable test code when technology changes again.

Application Scalability

The concept of scalability goes beyond just computer buses and platforms. Because every element in the digital measurement system is completely independent, anything is possible. For example, the following scenarios could all use the same software application:

- Research laboratory (flexibility is key) – Acquires and displays data from a digital multimeter (DMM) and oscilloscope connected via GPIB to a desktop computer.
- Field service application (portability is key) – Acquires and displays data from a computer-based DMM and oscilloscope plugged into a laptop via PCMCIA.
- Production test application (performance is key) – Acquires and displays data from a computer-based PXI/CompactPCI DMM and oscilloscope plugged into a PXI chassis.

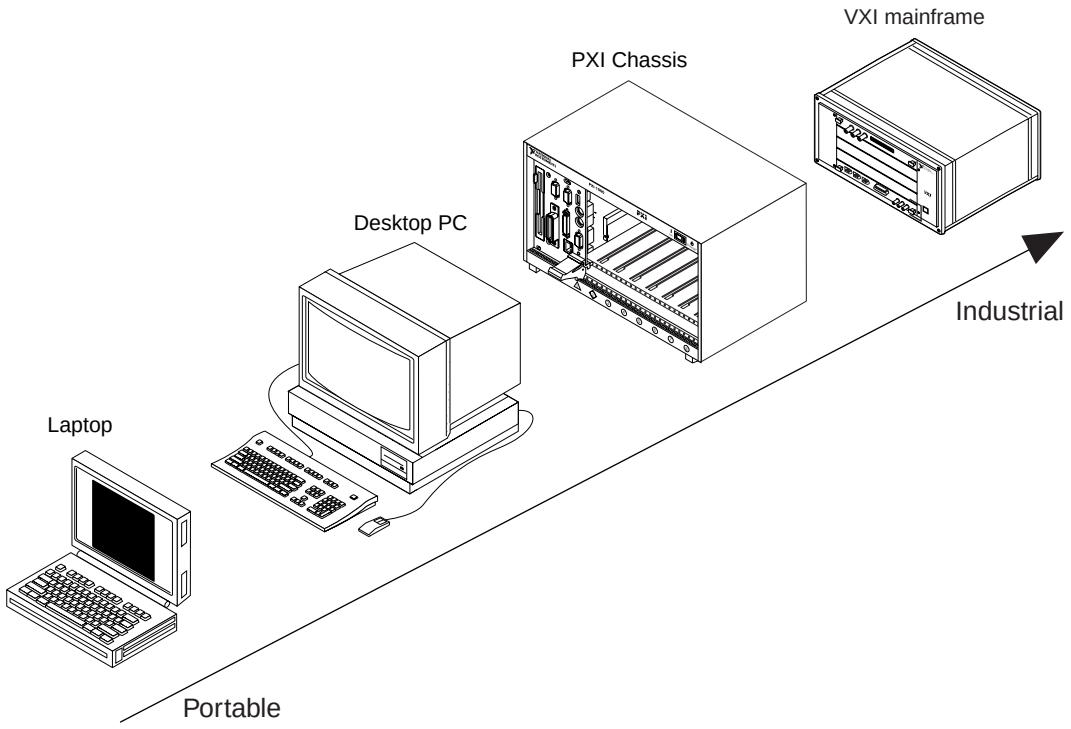


Figure 5. Applicatoin and Platform Scalability

The key is that the elements of a digital measurement system must be modular so the developer does not need to overhaul the entire system every time the computer industry extends a new technology to benefit measurements. For instance, just because a new operating system hits the scene, there's no need to upgrade all of the measurement hardware. Or, just because the latest computer bus offers better performance, there's no need to change the application software.

The Platform for the Future

The benefits of a digital measurement system are obvious:

- Increased performance
- Tighter systems integration
- Lower costs
- Better development productivity

Thus, it is a platform that continues to gain popularity.

As an outgrowth of the Measurement Revolution, the digital measurement system has no limits. It continues to unfold as computing technology evolves. For example, standard computing technologies are advancing so quickly that soon, embedded and real-time applications will be just an extension of the digital measurement system. The possibility of embedded and real-time monitoring control with standard computer technologies is perhaps greater than what we have seen today with computer-based measurement and automation systems.

Looking forward, there is every reason to believe that the computer will continue to be the driving technological factor in the advancement of measurement and automation systems. The enormous cost benefits of computer-based systems, coupled with their performance advantages will drive our industry for years to come.

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